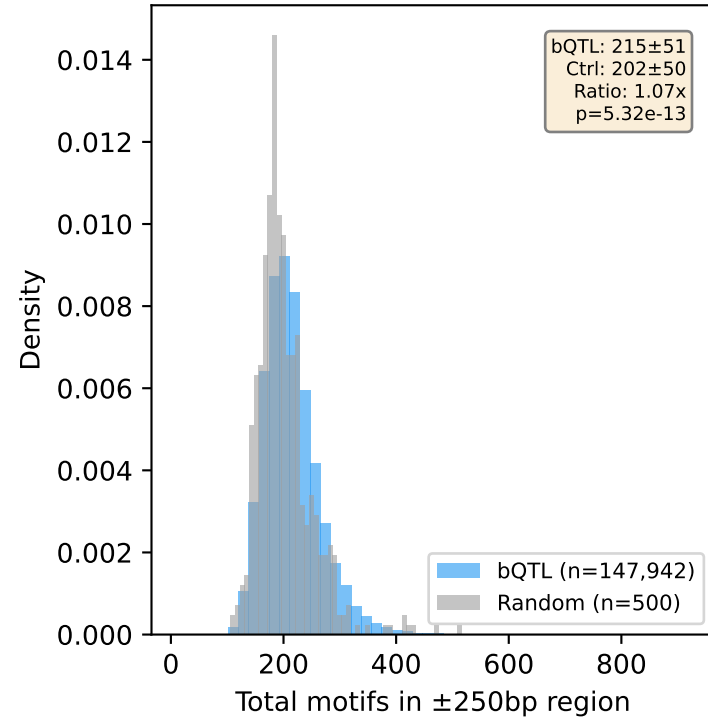
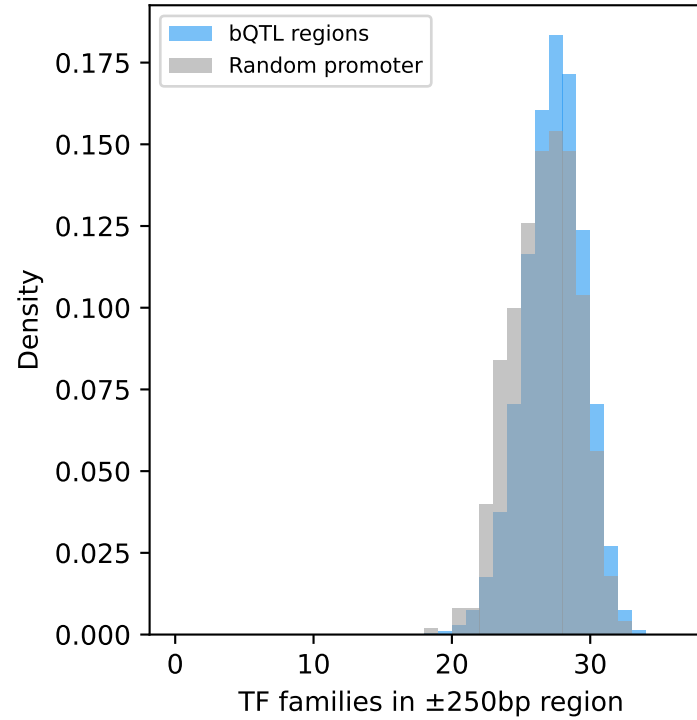


MOA-seq Bound Region Motif Analysis ($\pm 250\text{bp}$ around bQTL)

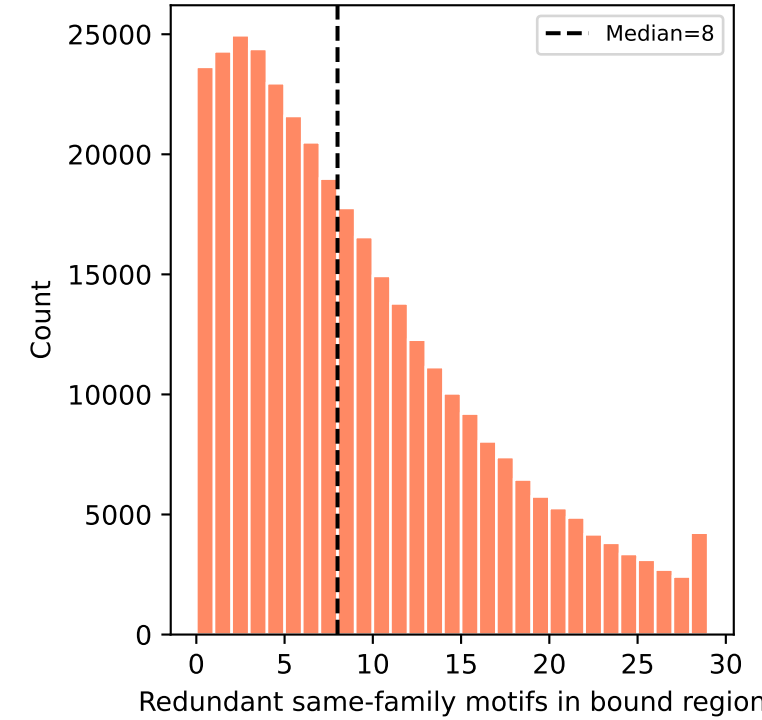
A. Motif density: bQTL vs random promoter



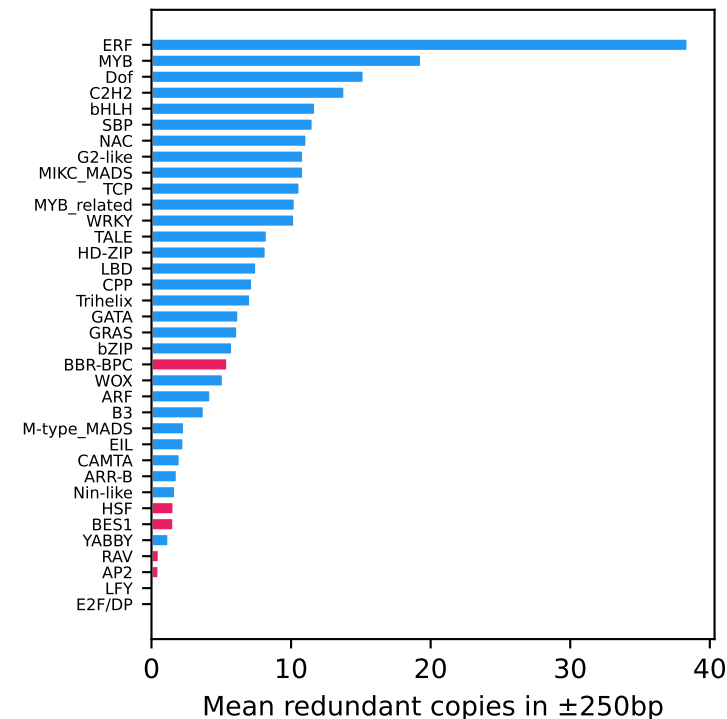
B. Family diversity: bQTL vs random



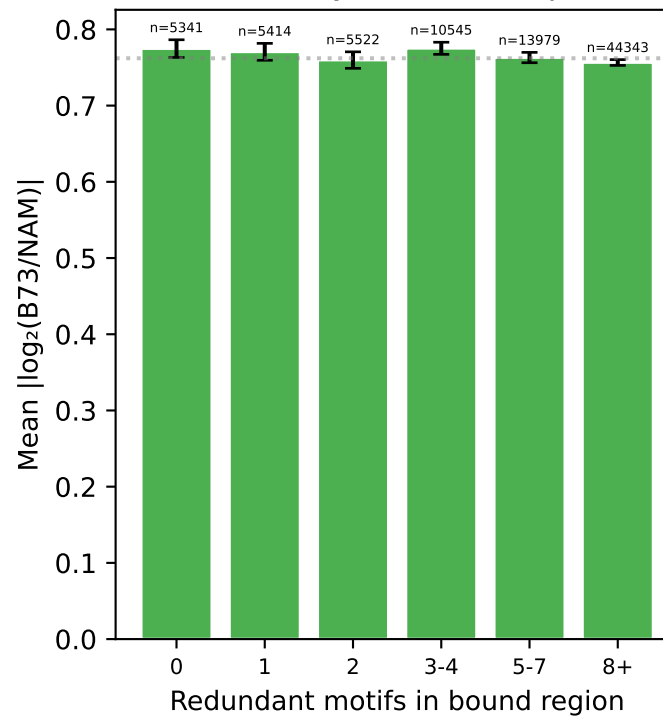
C. Within-region redundancy (sole: 6.3%)



D. Per-family redundancy (red = $>40\%$ sole)



E. Redundancy vs allelic expression



KEY FINDINGS

Motif density ($\pm 250\text{bp}$):
 bQTL regions: 215 motifs
 Random promoter: 202 motifs
 Ratio: 1.07x (near-identical)

91.5% of bQTL have TF motif
 at variant position

Within-region redundancy:
 Mean: 12 same-family copies
 Only 6.3% are sole motifs

Redundancy vs ASE:
 $\rho = -0.006$ (no correlation)
 Cohen's d = 0.016 (no effect)